WEEK 03 REPORT

THE CLEANING SPRINT

Project: Moneyball – IPL Player Valuation
Course: 24ADI204 – Data Science and Visualization
Event: Data-Viz Hack-A-Bit (2026–2027)
Team Number: 05

Team Member	Register Number
Divyesh D S	25BAD022
Kavi Prakash M	25BAD043
Mithuin P S	25BAD057
Madhan P M	25BAD503
Renganathan G M	25BAD087

Weekly project documentation based on the executed Week 03 Google Colab notebook.

1. Executive Summary

Week 03 focused on inspecting data quality, reviewing missing values, checking duplicate records and detecting potential outliers in the IPL Moneyball master dataset. A separate working copy was created so the original dataset remained unchanged. Missing-value patterns were visualized, boxplots were generated, and two statistical approaches—Interquartile Range (IQR) and Z-score—were used to flag potential outliers.

Key outcome: The dataset contains 3,311 rows and 27 columns. The inspection found 17,839 missing cells and zero duplicate rows. Potential outliers were identified, but no outlier rows were deleted. The missing-value total remained unchanged because the only fill operation targeted *bat_style*, which already had zero missing values.

2. Objectives

• Inspect missing values by column and percentage.
• Visualize missing-value patterns using a heatmap.
• Review categorical and numeric fields with missing values.
• Check duplicate records.
• Visualize selected numeric columns using boxplots.
• Detect potential outliers using IQR and Z-score methods.
• Compare the original dataset with the working copy and record the final data-quality status.

3. Dataset Profile

The notebook loaded the same project master file, **IPL_Moneyball_Merged_2008_2026.csv**, used in the earlier weeks. The file contains player-season information, batting and bowling statistics, auction-related details, team and player-profile attributes, and derived performance measures.

Attribute	Observed value
Dataset	IPL_Moneyball_Merged_2008_2026.csv
Rows	3,311
Columns	27
Total missing cells	17,839
Duplicate rows	0

4. Tools and Libraries

Pandas: CSV loading, DataFrame inspection, missing-value counts, duplicate checks and summary tables.
NumPy: numerical operations and absolute-value calculations.
Matplotlib and Seaborn: missing-value heatmap and boxplots.
SciPy: Z-score calculation for outlier detection.

5. Methodology

The CSV was loaded into a Pandas DataFrame and its shape was confirmed as 3,311 × 27. A working copy named **cleaned_df** was created using *df.copy()*. Missing counts and percentages were calculated for every column. A heatmap was plotted to show the pattern of missing entries. The notebook then inspected categorical and numeric columns separately, checked duplicates, plotted boxplots for selected statistics, calculated IQR bounds and counted Z-score outliers where |Z| > 3.

6. Missing-Value Analysis

The notebook reported the following columns with missing values:

Column	Missing	Percentage
base_price_lakh	3,109	93.90%
role	2,418	73.03%
nationality	2,296	69.34%
bowling_avg	1,520	45.91%
final_price_lakh	1,480	44.70%
status	1,200	36.24%
team	1,200	36.24%
auction_name	1,200	36.24%
acquisition_type	1,200	36.24%
economy	1,130	34.13%
batting_avg	677	20.45%
strike_rate	383	11.57%
bowl_style	26	0.79%

The notebook also showed that player, bat_style and player_full_name had no missing values. The heatmap provided a visual overview of missingness across records and columns.

Handling decision recorded in the notebook: *bat_style* was filled with the label “Unknown”. However, its missing count was already zero, so this operation did not change the dataset. Other missing fields were retained rather than being filled without a verified meaning for the missing entries.

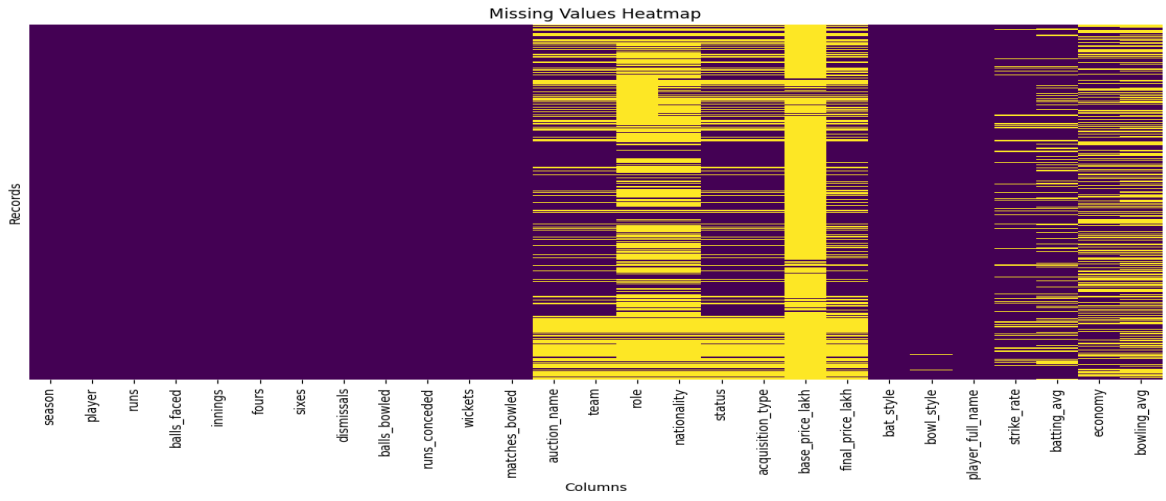
7. Duplicate-Record Check

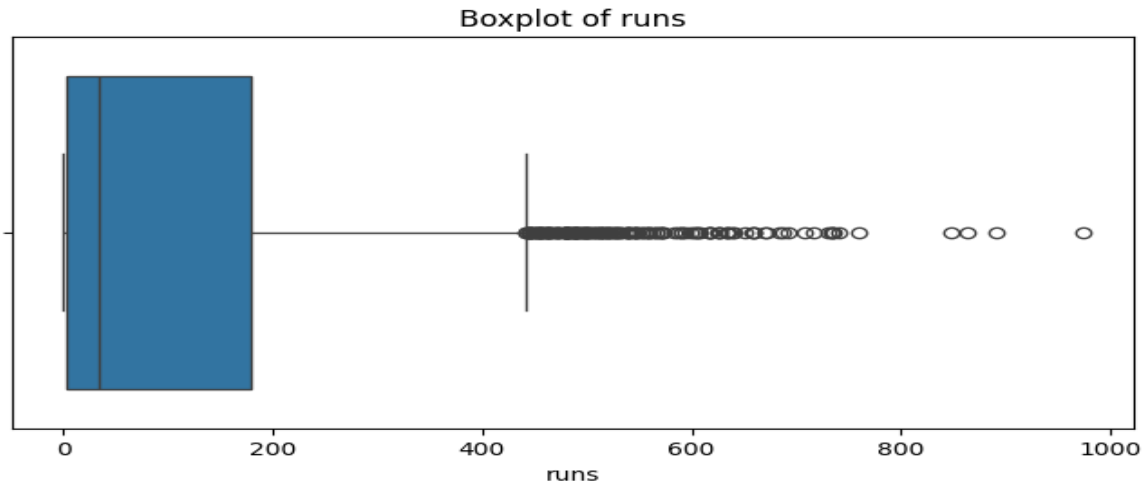
The duplicate check returned **0 duplicate records**. Therefore, no duplicate rows were removed from the working copy.

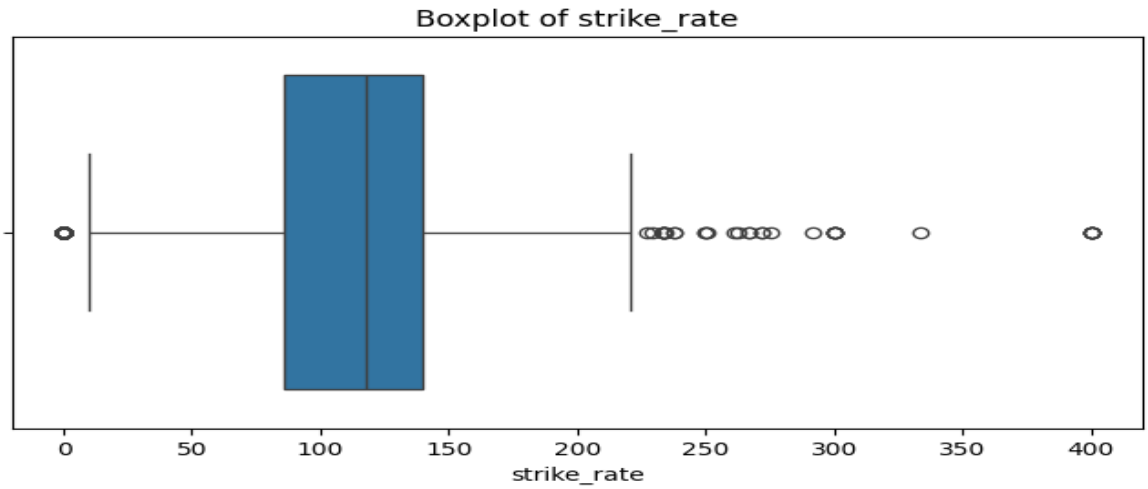
8. Boxplot-Based Visual Inspection

Boxplots were generated for runs, strike_rate, batting_avg, wickets, economy and final_price_lakh. The plots showed points beyond the whiskers in several distributions. These points were treated as potential outliers for further statistical checking, not as automatic data errors.

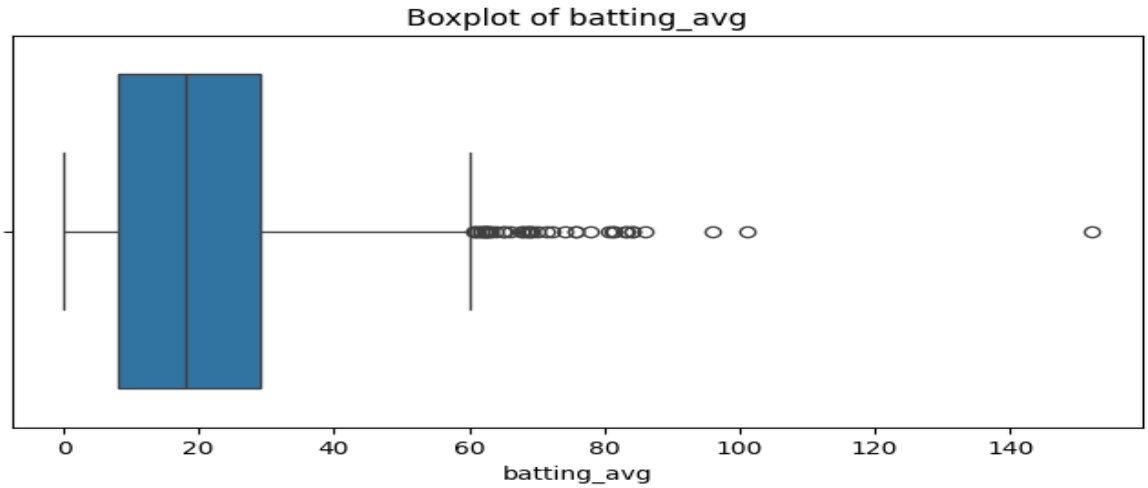
Boxplots generated by the executed notebook

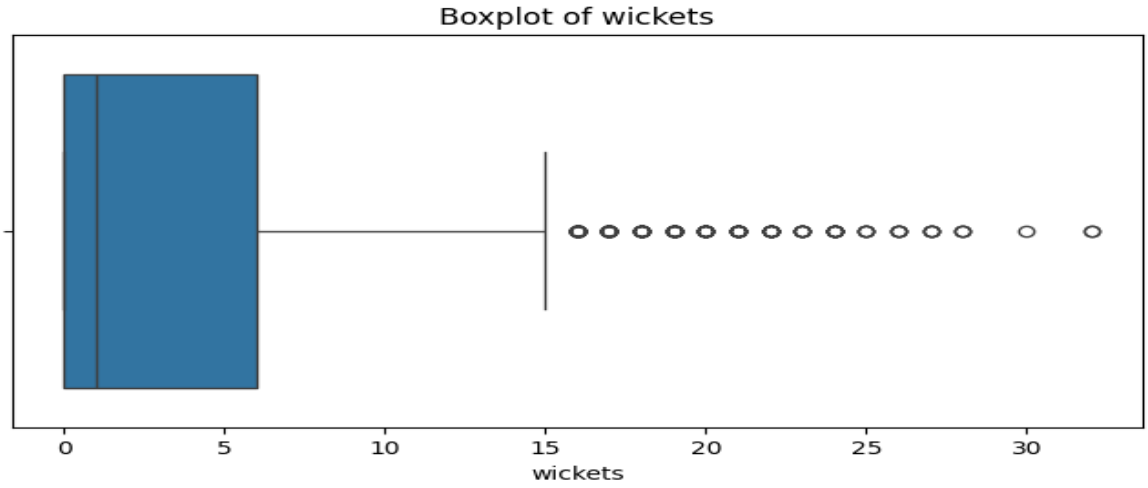


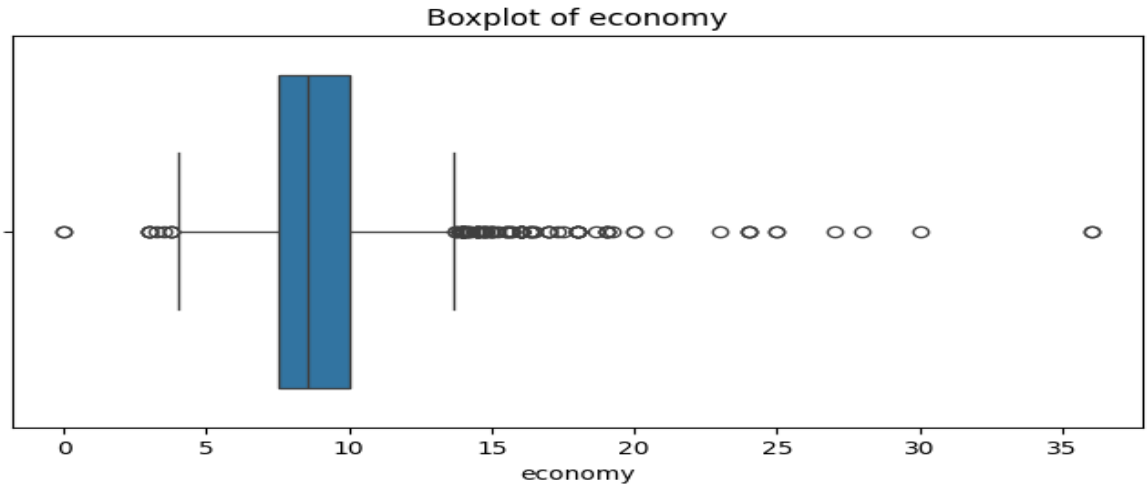

Boxplot of runs
0
200
400
600
800
1000
runs

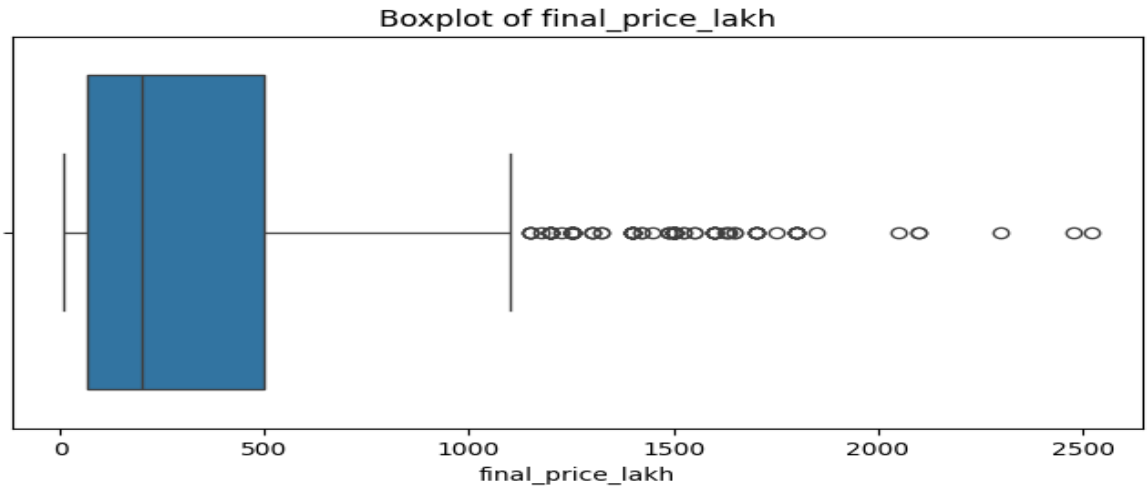

Boxplot of strike_rate
0
50
100
150
200
250
300
350
400
strike_rate

8. Boxplot-Based Visual Inspection (continued)









9. IQR Outlier Detection

The IQR method used Q1 and Q3 to calculate the interquartile range, followed by lower and upper fences using Q1 – 1.5 × IQR and Q3 + 1.5 × IQR. The counts below are the actual notebook outputs.

Column	Lower limit	Upper limit	Outliers
runs	-258.500	441.500	156
strike_rate	4.275	221.435	182
batting_avg	-23.500	60.500	39
wickets	-9.000	15.000	203
economy	3.825	13.705	107
final_price_lakh	-578.125	1,146.875	130

Among these six columns, wickets had the largest IQR outlier count (203), while batting_avg had the smallest (39). The negative lower fences are mathematical thresholds; they do not indicate that negative observed runs, wickets or prices were present.

10. Z-score Outlier Detection

The notebook calculated Z-scores for five numeric columns and flagged values with an absolute Z-score greater than 3.

Column	Z-score outliers
runs	46
strike_rate	20
batting_avg	25
wickets	44
economy	31

Z-score was not applied to final_price_lakh in the executed notebook. The results from IQR and Z-score are not expected to match because the methods use different statistical criteria and distribution assumptions.

11. Comparison of IQR and Z-score

Column	IQR	Z-score
runs	156	46
strike_rate	182	20
batting_avg	39	25
wickets	203	44
economy	107	31

Across the five columns included in the comparison table, IQR flagged 687 column-level outlier instances and Z-score flagged 166. These are sums of counts by column, not counts of unique rows; a row may be flagged in more than one column.

12. Before-and-After Dataset Comparison

Metric	Original	Working copy
Rows	3,311	3,311
Columns	27	27
Missing values	17,839	17,839
Duplicate records	0	0

The original and working copy have identical dimensions, missing-value totals and duplicate counts. This confirms that the notebook did not delete rows or reduce missingness. The bat_style fill operation had no effect because the column already contained zero missing values.

13. Final Findings and Conclusion

Week 03 successfully completed a data-quality inspection and outlier-detection workflow. The team confirmed the dataset size, reviewed missingness, visualized missing-value patterns, checked duplicates and applied IQR and Z-score methods to identify potential outliers. The analysis found 17,839 missing cells and no duplicate records. Several performance and auction-related columns contain potential outliers, but these values were retained because extreme IPL statistics or auction prices may be genuine observations.

This week should be described as a cleaning assessment and detection stage rather than completed full data cleaning: no missing values were actually imputed, no rows were removed, and no outliers were treated. These results provide a documented baseline for any later transformation decisions.

14. Output File and GitHub Structure

The notebook executed a CSV export command for **IPL_Moneyball_Cleaned_Week03.csv** and printed “Dataset saved successfully!”. The Week 03 GitHub folder should contain exactly two files:

Week03_Cleaning_Sprint.ipynb – executed notebook with code, outputs and visualizations.
Week03_Report.pdf – this report documenting the process and findings.

15. References

1. Team master dataset: IPL_Moneyball_Merged_2008_2026.csv.
2. Executed project notebook: Week03_Cleaning_Sprint.ipynb.
3. Data-Viz Hack-A-Bit (2026–2027), Data Science and Visualization, course code 24ADI204 – project weekly workflow.